

# EE445 Mod2-Lec3: Least Squares Regression

## References:

- [VMLS]: Chapter 12

# Agenda

- Optimality via linear algebra (eigenvalues of Hessian & linear independence)
- Least squares solution via normal equations

# Linear Models: Predicting Outputs from Inputs

- **Goal:** learn a function (aka "model")  $f_{\theta}(x)$  that predicts quantity  $y \in \mathbb{R}$  from input  $x \in \mathbb{R}^d$ .
- For linear models:

$$f_{\theta}(x) = \theta^{\top} x = \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d \quad (\text{linear})$$

$$f_{\theta}(x) = \theta^{\top} \begin{bmatrix} x \\ 1 \end{bmatrix} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d \quad (\text{affine})$$

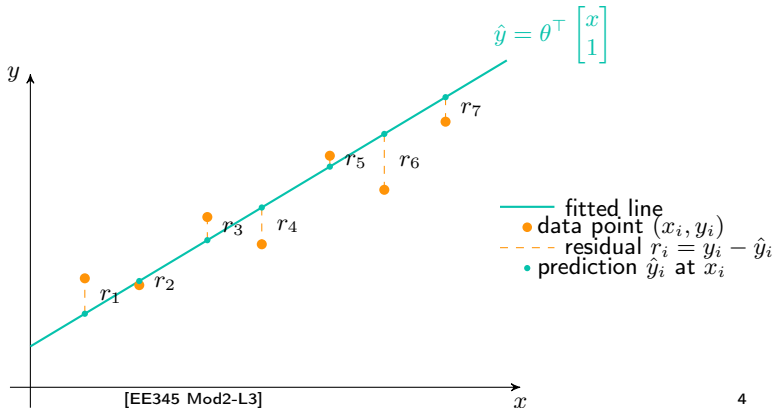
- Given data  $(x_i, y_i)$  for  $i = 1, \dots, n$ , we choose  $\theta$  to minimize prediction error.
- Examples:
  - ▶ Predict house price from size, bedrooms, age.
  - ▶ Predict exam score from study hours.

# Measuring Predictor Error in 2D (Linear Regression)

For a fitted line  $\hat{y} = \theta_1 x + \theta_0$  and points  $(x_i, y_i)$ , the **residual** is

$$r_i = y_i - \hat{y}_i = y_i - \theta x_i - \theta_0, \quad \text{and} \quad \mathcal{L}(\theta_1, \theta_0) = \sum_{i=1}^n r_i^2.$$

**OLS loss:** minimize total squared residual length  
 $\sum_i \|r_i\|_2^2$  (vertical errors).



# Empirical Risk Minimization (ERM): Quintentessential ML Problem

**Goal of supervised learning:** Find a function that makes good predictions on new data.

**Challenge:** We don't know the true data distribution; indeed, we only see a **finite** set of samples.

**Idea of ERM:**

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(x_i), y_i) \quad \text{average over } n \text{ samples} = \text{empirical risk}$$

# Least Squares (Linear Regression) Set-up

- Consider an  $n \times d$  matrix  $X \in \mathbb{R}^{n \times d}$  and vectors  $y \in \mathbb{R}^n$ ,  $\theta \in \mathbb{R}^d$

## Least Squares Optimization Problem

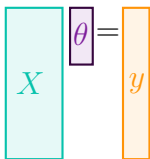
Find  $\theta$  such that  $X\theta = y$  that minimizes the  $\ell_2$  loss:

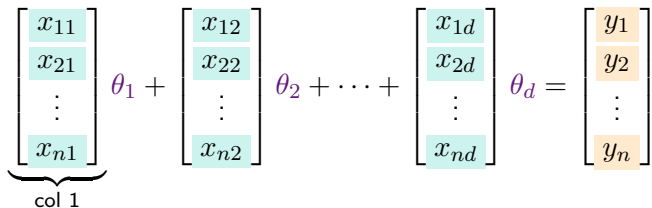
$$\min_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta) = \min_{\theta \in \mathbb{R}^d} \|y - X\theta\|_2^2$$

- $X$  is a **matrix of training data**; e.g.,  $n$  is number of samples,  $d$  is number of 'features'
- $y \in \mathbb{R}^n$  contains **'target values', 'labels', or observations of real world phenomena**
- $\theta \in \mathbb{R}^d$  is a set of **feature weights**

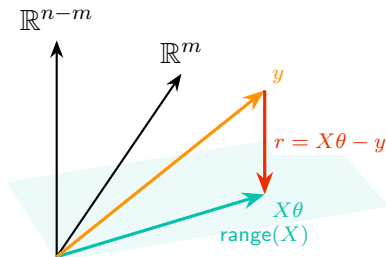
# Overdetermined System of Equations $n \gg d$

- **Goal:** Find a solution to  $X\theta = y$ —that is, find  $\theta$  such that  $X\theta = y$
- **Overdetermined System:**  $X$  is a 'tall' matrix  $\implies$  there are more samples ( $n$ ) than features to choose ( $d$ ).


$$X\theta = y$$


$$\underbrace{\begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{bmatrix}}_{\text{col 1}} \theta_1 + \begin{bmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{bmatrix} \theta_2 + \cdots + \begin{bmatrix} x_{1d} \\ x_{2d} \\ \vdots \\ x_{nd} \end{bmatrix} \theta_d = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Projection Geometry:  $y = X\theta + r$  with  $r \perp \text{range}(X)$



$$m = \dim(\text{range}(X))$$



# Finding Minima and Convexity

**Goal:** Find  $\theta^*$  that minimizes a differentiable function  $f : \mathbb{R}^d \rightarrow \mathbb{R}$ .

Conditions for Minima [VMLS, App. C]

first order condition:  $\nabla f(\theta^*) = 0$  (stationary point)

Second-order condition :  $v^\top \underbrace{\nabla^2 f(\theta^*)}_{=: H(\theta^*)} v > 0$  for all  $v \neq 0$ , (convexity)

- Gradient = direction of steepest ascent.
- Setting it to zero  $\implies$  flat point.
- Positive-definite Hessian  $\implies$  surface curves upward in all directions.
- $H(\theta) = \nabla^2 f(\theta)$  is the Hessian matrix.
- we say  $f$  is **strictly convex** near  $\theta^*$
- $\theta^*$  is the unique minimum.

# Example: Minima in Two Dimensions

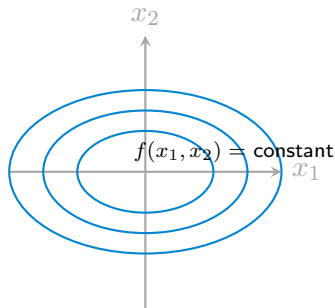
Now  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  has a gradient and Hessian:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}, \quad H = \nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}.$$

**Example:**  $f(x_1, x_2) = ax_1^2 + bx_2^2$  for  $a, b > 0$ .

$$H = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \implies v^\top H v = av_1^2 + bv_2^2.$$

$\implies H \succ 0$  (i.e. positive definite) if  $a > 0$  and  $b > 0$ .



# A More General 2D Hessian: Tilted Ellipses

**Example (coupled quadratic):**

$$f(x_1, x_2) = \frac{1}{2} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \underbrace{\begin{bmatrix} a & c \\ c & b \end{bmatrix}}_H \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (a, b > 0).$$

**Gradient & Hessian:**

$$\nabla f(x) = H x, \quad \nabla^2 f(x) = H = \begin{bmatrix} a & c \\ c & b \end{bmatrix}.$$

# General Convexity Condition via Eigenvalues

For any symmetric Hessian  $H \in \mathbb{R}^{d \times d}$ ,

$$H \text{ is positive definite} \iff v^\top H v > 0 \text{ for all } v \neq 0.$$

Checking this directly is hard, so we use **eigenvalues**.

# General Convexity Condition via Eigenvalues

## Diagonalizing Symmetric Matrices (Spectral Theorem)

For general symmetric matrix  $H$ , there exists  $Q$  such that  $QQ^\top = I$  and  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$  such that  $H = Q\Lambda Q^\top$ .

**What we will show:**

# Linear Algebra Concepts

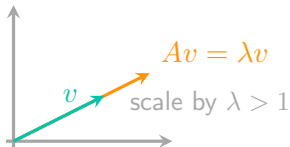
- Eigenvalues and eigenvectors
- Linear independence of eigenvectors
- Basis of eigenvectors
- Diagonalizing symmetric matrices
- Spectral theorem
- Positive definiteness

# Eigenvalues and Eigenvectors

## Definition of Eigenvalues and Eigenvectors

### Geometric meaning:

- $A$  scales (stretches or flips)  $v$  without changing its direction.
- The eigenvalues describe how much  $A$  stretches space along special directions.
- If  $\lambda = 0$ , that direction is *flattened*  $\rightarrow$  indicates "rank deficiency".



# Independence of Eigenvectors and Basis Formation

## Linear Independence



# Examples: Checking Linear Independence

## Example 1

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Check if  $\{v_1, v_2, v_3\}$  are linearly independent in  $\mathbb{R}^2$ .

# Examples: Checking Linear Independence

## Example 2

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

# Independence of Eigenvectors and Basis Formation

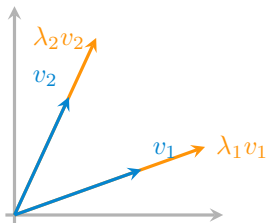
## Theorem

If  $Av_i = \lambda_i v_i$  and  $Av_j = \lambda_j v_j$  with  $\lambda_i \neq \lambda_j$ , then  $v_i$  and  $v_j$  are linearly independent.

# Independence of Eigenvectors and Basis Formation

## Facts

- For  $A \in \mathbb{R}^{d \times d}$  with  $d$  distinct eigenvalues, its eigenvectors  $v_1, \dots, v_d$  are linearly independent.
- If  $A$  has  $d$  **distinct** eigenvalues, its eigenvectors form a **basis** for  $\mathbb{R}^d$ .
- Distinct eigenvalues guarantee **independent** eigenvectors, which form a basis that **diagonalizes**  $A$ .



# What is a Basis?

## Definition of Basis

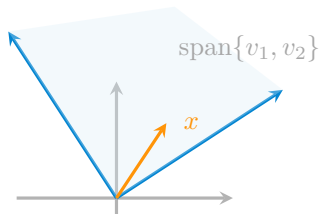
A *basis* of a vector space  $V \subseteq \mathbb{R}^d$  is a set of vectors  $\{v_1, v_2, \dots, v_d\}$  such that:

(1)  $v_1, \dots, v_d$  are linearly independent,      (2) they span  $V$ .

(2)  $\iff$  every vector  $x \in V$  can be written *uniquely* as  $x = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_d v_d$ .

### Key Ideas:

- “Span”  $\implies$  the basis vectors *reach* all of  $V$ .
- “Linear independence”  $\implies$  no vector in the set is redundant.
- The number of basis vectors = **dimension** of  $V$ .



# Independence of Eigenvectors and Basis Formation

## Why this matters:

- If  $A \in \mathbb{R}^{d \times d}$  has  $d$  linearly independent eigenvectors, they form a **basis** for  $\mathbb{R}^d$ .
- In that basis,  $A$  acts simply by scaling each basis vector:

$$Av_i = \lambda_i v_i.$$

## Diagonalization

# Where are we

- We have defined linear independence, and even shown how to diagonalize a matrix if it has independent eigenvectors.
- Now we want to come back to this Hessian matrix and questions about curvature.
- To apply the eigenvalue check, we need to show that Hessians can be decomposed as  $H = A^\top A$  for some  $A$ .

# Why Symmetric Matrices Can Be Diagonalized

## Claim (Spectral Theorem)

If  $H \in \mathbb{R}^{d \times d}$  is **symmetric** ( $H = H^\top$ ), then there exists an **orthogonal matrix**  $Q$  (i.e.,  $Q^\top Q = I$ ) and a diagonal matrix  $\Lambda$  such that

$$H = Q\Lambda Q^\top, \quad \text{where } \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d), \text{ and } Hq_i = \lambda_i q_i.$$



# Why Eigenvectors of a Symmetric Matrix Are Orthogonal?

## What we have shown so far...

- Optimality:  $\nabla f(\theta^*) = 0$  and  $H(\theta^*) \succ 0 \implies \theta^*$  is a (local) minimum.

$$H(\theta) = \nabla^2 f(\theta) = \begin{bmatrix} \frac{\partial^2 f}{\partial \theta_1^2} & \cdots & \frac{\partial^2 f}{\partial \theta_1 \partial \theta_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial \theta_d \partial \theta_1} & \cdots & \frac{\partial^2 f}{\partial \theta_d^2} \end{bmatrix} \quad (\text{Hessian})$$

$H(\theta) \succ 0 \iff v^\top H(\theta) v > 0 \quad \forall v \neq 0 \implies f$  is **strictly convex** at that point

- **Eigenvalue test:** Since  $H$  is symmetric, we have that

$$H = Q\Lambda Q^\top, \quad Q^\top Q = I, \quad \text{where } \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$$

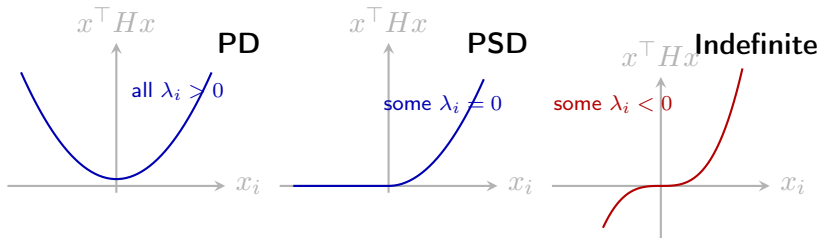
- This means that  $H$  is positive definite if and only if all its eigenvalues are positive:  
 $H \succ 0 \iff$  all eigenvalues  $\lambda_i > 0$ .

# Eigenvalues and Curvature: PD, PSD, and Indefinite

For a symmetric matrix  $H$ , curvature along direction  $v$  is measured by

$$v^{\top} H v = \lambda \|v\|_2^2.$$

- $\lambda_i > 0 \implies$  curves upward (bowl) • curvature in every direction  $\rightarrow$  unique minimum.
- $\lambda_i = 0 \implies$  flat direction (ridge) • flat direction  $\rightarrow$  multiple minima or constraints.
- $\lambda_i < 0 \implies$  curves downward (saddle) • saddle or concave directions  $\rightarrow$  not convex.



# Computing Eigenvalues: The Characteristic Polynomial

## Definition of Eigenvalues and Eigenvectors

# Why $\det(A - \lambda I) = 0$ Defines Eigenvalues

Eigenvalue equation

Fact

$\lambda$  is an eigenvalue of  $A \iff \det(A - \lambda I) = 0$ .

# Why $\det(A - \lambda I) = 0$ Defines the Eigenvalues

## Example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

## Takeaway

Eigenvalues are the roots of the polynomial  $\det(A - \lambda I) = 0$ .

## 3×3 Example

### Example

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}. \quad p_A(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 2 - \lambda & 0 & 0 \\ 0 & 2 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{bmatrix}.$$

# $3 \times 3$ Example: Eigendecomposition



# Eigenvalues in Higher Dimensions: The Characteristic Polynomial

Characteristic polynomial

# Calculus & Linear Algebra Summary

- **Optimality conditions:**  $\nabla f(\theta^*) = 0$  and  $H(\theta^*) \succ 0 \implies \theta^*$  is a (local) minimum.
- **Symmetric matrices can be diagonalized:**  $H = Q\Lambda Q^\top$  for some orthogonal matrix  $Q$  and diagonal matrix  $\Lambda$  with eigenvalues  $\lambda_1, \dots, \lambda_d$ .
- **Positive definiteness:**  $H \succ 0 \iff$  all eigenvalues  $\lambda_i > 0$ .
- **Computing eigenvalues:** Eigenvalues are the roots of the characteristic polynomial  $\det(A - \lambda I) = 0$ .

# Solving Least Squares (Coordinate Form)

For one data point:  $f_{\theta}(x_i) = \theta_1 x_{i1} + \theta_2 x_{i2} + \cdots + \theta_d x_{id} = \theta^{\top} x$ .

$$\text{minimize } \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta_1 x_{i1} - \theta_2 x_{i2} - \cdots - \theta_d x_{id})^2 = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n x_{ij} x$$

# Solving Least Squares (Coordinate Form)

# Matrix Derivatives of Quadratic Forms

**Function:**  $f(x) = x^\top Ax$ , where  $A \in \mathbb{R}^{n \times n}$ .

**Step 1. Expand in coordinates:**

$$f(x) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j.$$

Then the partial derivative with respect to  $x_k$  is

$$\frac{\partial f}{\partial x_k} = \sum_{j=1}^n a_{kj} x_j + \sum_{i=1}^n a_{ik} x_i.$$

**Combine in vector form:**

$$\nabla f = (A + A^\top)x.$$

If  $A$  is symmetric,  $A = A^\top$ , so:

$$\nabla(x^\top Ax) = 2Ax.$$

# Solving Least Squares (Vector Form)

Recall the objective:

$$\mathcal{L}(\theta) = \frac{1}{n} \|y - X\theta\|_2^2 = \frac{1}{n} (y - X\theta)^\top (y - X\theta).$$

# Solving Least Squares (Vector Form)

Recall the objective:

$$\mathcal{L}(\theta) = \frac{1}{n} \|y - X\theta\|_2^2 = \frac{1}{n} (y - X\theta)^\top (y - X\theta).$$

# Coordinate vs. Vector Gradient (2D Example)

Consider two features:  $x_i = (x_{i1}, x_{i2})$  and parameters  $\theta = (\theta_1, \theta_2)$ .

$$\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - x_{i1}\theta_1 - x_{i2}\theta_2)^2 = \frac{1}{n} \|y - X\theta\|_2^2.$$

**Coordinate form:**

$$\frac{\partial \mathcal{L}}{\partial \theta_1} = -\frac{2}{n} \sum_{i=1}^n x_{i1}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2),$$

$$\frac{\partial \mathcal{L}}{\partial \theta_2} = -\frac{2}{n} \sum_{i=1}^n x_{i2}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2).$$

**Vector form:**

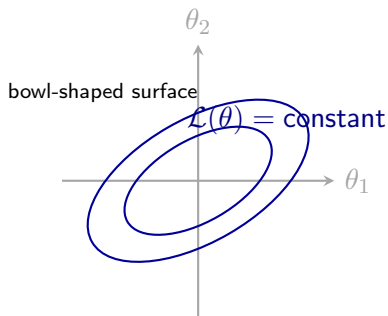
$$\begin{aligned} \nabla_{\theta} \mathcal{L}(\theta) &= -\frac{2}{n} X^{\top} (y - X\theta) \\ &= -\frac{2}{n} \begin{bmatrix} \sum_i x_{i1}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2) \\ \sum_i x_{i2}(y_i - x_{i1}\theta_1 - x_{i2}\theta_2) \end{bmatrix}. \end{aligned}$$



# Convexity of the Least Squares Objective

$$\mathcal{L}(\theta) = \frac{1}{n} \|y - X\theta\|_2^2 = \frac{1}{n} (y - X\theta)^\top (y - X\theta).$$

# Convexity Intuition (2D Example)



**Takeaway:** Least squares is convex because its Hessian  $\frac{2}{n}X^\top X$  is positive definite when  $X$  has full column rank.

What happens if some directions have zero curvature?

# Back to the optimization problem...

What does it mean to be a solution?

# Column Interpretation

$$X = \begin{bmatrix} | & \cdots & | \\ x_1 & \cdots & x_d \\ | & \cdots & | \end{bmatrix}, \quad x_i \in \mathbb{R}^n$$

# Row Interpretation

$$X = \begin{bmatrix} - & \tilde{x}_1^\top & - \\ \vdots & \vdots & \vdots \\ - & \tilde{x}_n^\top & - \end{bmatrix}, \quad \tilde{x}_i \in \mathbb{R}^d$$

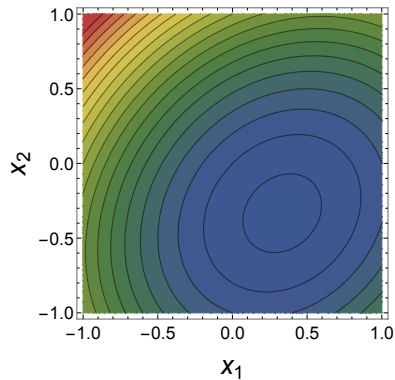
## 3×2 Example

### Example

$$X = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad n = 3, \quad d = 2$$

## Example Continued

$$\min_{\theta_1, \theta_2} \{ (2\theta_1 - 1)^2 + (-x_1 + \theta_2)^2 + (2\theta_2 + 1)^2 \}$$



# Least Squares Solution Properties & Facts

## Assumption [A1]:

The columns of  $X$  are linearly independent; i.e.,  $\sum_{i=1}^n c_i x_i = 0 \iff c_i = 0 \quad \forall i = 1, \dots, n$



# Why "Pseudo-Inverse"?

For a square invertible matrix  $A$ , the inverse  $A^{-1}$  satisfies  $A^{-1}A = I$ .

Why "Pseudo-Inverse"?

# The Pseudo-Inverse is a Projection Matrix

## The Pseudo-Inverse is a Projection Matrix

The matrix  $P = X(X^\top X)^{-1}X^\top$  is the orthogonal projection onto the column space of  $X$ .

Properties:

- $P^2 = P$  (idempotent): Projecting twice = projecting once
- $P^\top = P$  (symmetric): Orthogonal projection
- $Py$  is the closest point in  $\text{col}(X)$  to  $y$
- $(I - P)$  projects onto the orthogonal complement (residuals in regression)

In linear regression  $\hat{y} = X\hat{\theta} = X(X^\top X)^{-1}X^\top y = Py$  is the projection of  $y$  onto the column space.

# Direct Verification that the Solution is a Minimum

## Example Problem: Solution is a Minimum

Show that for any  $\theta \neq \hat{\theta} = X^\dagger y$  we have

$$\|X\hat{\theta} - y\|_2^2 < \|X\theta - y\|_2^2$$

# Direct Verification that the Solution is a Minimum

# Projection

- The least squares solution is  $\hat{\theta} = (X^\top X)^{-1} X^\top y$  and the prediction is  $\hat{y} = X\hat{\theta}$
- $P = X(X^\top X)^{-1} X^\top$  is a projection matrix: it projects onto the subspace formed by the columns of  $X$ 
  - ▶  $P$  is a projection matrix if  $P^2 = P$
- Orthogonal decomposition:

$$y = y_{\mathcal{R}(X)} + y_{\mathcal{R}(X)^\perp} \quad \text{where} \quad y_{\mathcal{R}(X)} = X\hat{\theta} \quad \text{and} \quad y_{\mathcal{R}(X)^\perp} = r = X\hat{\theta} - y$$

# Example

## Example: Over-determined System

Consider

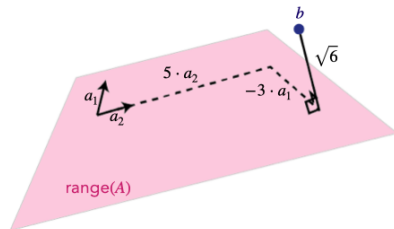
$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

Find the least squares approximate solution to  $Ax = b$ .

## Example Continued

- The solution minimizes the distance from  $A\hat{x}$  to  $b$ :

$$\|y - X\hat{\theta}\|_2^2 = \left\| \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\|_2^2$$



What about under-determined systems?



# Minimum Norm Solution

# Proof of Minimal Norm Property.

# Null Space Connection

# Pseudo-Inverse and Projection Formulas

# Next time

## Today we covered:

- Convexity, Hessians, and curvature
- Linear independence, basis, and dimension
- Why symmetric matrices can be diagonalized
- Checking convexity with the eigenvalue test
- Solving least squares by setting the gradient to zero

## Next time we will cover:

- Under-determined systems
- (next week) regularization and ridge regression
- (next week) bias-variance tradeoff, and generalization error